

BÁO CÁO BÀI TẬP VỀ NHÀ 01

Phan Huy Hoàng – 20235729

1. HelloWorld.java: Yêu cầu người dùng nhập tên của mình, sau đó hiển thị lời chào đến người đó.

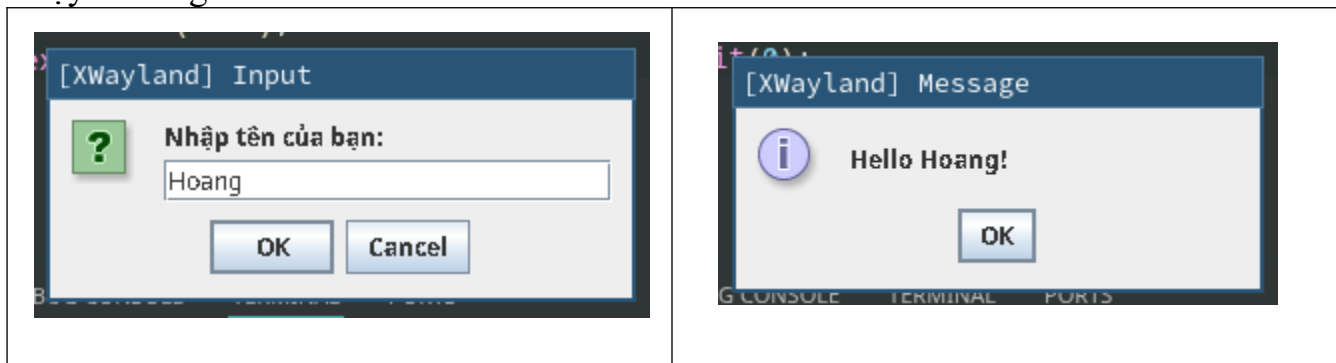
Code:

```
import javax.swing.*;

public class HelloWorld {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Hello World");
        String name = JOptionPane.showInputDialog(frame, "Nhập tên của bạn:");

        JOptionPane.showMessageDialog(frame, "Hello " + name + "!");
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
        System.exit(0);
    }
}
```

Chạy chương trình:



2. Cài đặt chương trình Calculator, sử dụng mô hình MVC với 2 phiên bản:

a. Swing

Model:

```
package swing;

public class CalculatorModel {
    private float calculationValue;

    public void calculate(float firstNumber, float secondNumber, char operator) {
        switch (operator) {
            case '+':
                calculationValue = firstNumber + secondNumber;
                break;
            case '-':
```

```

        calculationValue = firstNumber - secondNumber;
        break;
    case '*':
        calculationValue = firstNumber * secondNumber;
        break;
    case '/':
        if (secondNumber != 0) {
            calculationValue = firstNumber / secondNumber;
        } else {
            throw new ArithmeticException("Cannot divide by
zero");
        }
        break;
    default:
        throw new IllegalArgumentException("Invalid operator");
    }
}

public float getCalculationValue() {
    return calculationValue;
}
}

```

View:

```

package swing;

import javax.swing.*;
import java.awt.BorderLayout;
import java.awt.GridLayout;
import java.awt.event.ActionListener;

public class CalculatorView extends JFrame {
    private JTextField firstNumber, secondNumber, calcSolution;
    private JButton addButton, subButton, mulButton, divButton;

    public CalculatorView() {
        setTitle("MVC Calculator");
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setSize(400, 200);

        firstNumber = new JTextField(10);
        secondNumber = new JTextField(10);
        calcSolution = new JTextField(10);
        calcSolution.setEditable(false);

        addButton = new JButton("+");
        subButton = new JButton("-");
        mulButton = new JButton("*");
    }
}

```

```

divButton = new JButton("/");

JPanel panel = new JPanel(new BorderLayout(10, 0));

JPanel infoPanel = new JPanel(new GridLayout(3, 2, 10, 10));
infoPanel.add(new JLabel("First Number:"));
infoPanel.add(firstNumber);
infoPanel.add(new JLabel("Second Number:"));
infoPanel.add(secondNumber);
infoPanel.add(new JLabel("Result:"));
infoPanel.add(calcSolution);

JPanel operandPanel = new JPanel(new GridLayout(1, 4, 10, 10));
operandPanel.add(addButton);
operandPanel.add(subButton);
operandPanel.add(mulButton);
operandPanel.add(divButton);

panel.add(infoPanel, BorderLayout.CENTER);
panel.add(operandPanel, BorderLayout.SOUTH);

getContentPane().add(panel);
}

public void addActionListener(ActionListener listener) {
    addButton.addActionListener(listener);
    subButton.addActionListener(listener);
    mulButton.addActionListener(listener);
    divButton.addActionListener(listener);
}

public String getFirstNumber() {
    return firstNumber.getText();
}

public String getSecondNumber() {
    return secondNumber.getText();
}

public void setCalcSolution(String solution) {
    calcSolution.setText(solution);
}
}

```

Controller:

```
package swing;
```

```
import java.awt.event.ActionEvent;
```

```

import java.awt.event.ActionListener;

public class CalculatorController implements ActionListener {
    private CalculatorModel model;
    private CalculatorView view;

    public CalculatorController(CalculatorModel model, CalculatorView view) {
        this.model = model;
        this.view = view;
        view.addActionListener(this);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        try {
            float firstNumber = Float.parseFloat(view.getFirstNumber());
            float secondNumber = Float.parseFloat(view.getSecondNumber());
            char operator = getOperator(e.getActionCommand());

            model.calculate(firstNumber, secondNumber, operator);
            view.setCalcSolution(String.valueOf(model.getCalculationValue()));
        } catch (NumberFormatException ex) {
            view.setCalcSolution("Error");
        } catch (ArithmeticException ex) {
            view.setCalcSolution("Error: Division by zero");
        } catch (IllegalArgumentException ex) {
            view.setCalcSolution("Error: Invalid operator");
        }
    }

    private char getOperator(String actionCommand) {
        switch (actionCommand) {
            case "+": return '+';
            case "-": return '-';
            case "*": return '*';
            case "/": return '/';
            default: throw new IllegalArgumentException("Invalid operator");
        }
    }
}

```

Driver:

```

package swing;

public class MVCCalculator {
    public static void main(String[] args) {
        CalculatorModel model = new CalculatorModel();
        CalculatorView view = new CalculatorView();
        CalculatorController controller = new CalculatorController(model,
view);
    }
}

```

```

        view.setVisible(true);
    }
}

```

Kết quả:

First Number:	1	First Number:	1
Second Number:	3	Second Number:	0
Result:	4.0	Result:	Error: Division by zero
+ - * /		+ - * /	

b. JavaFX:

Model:

```
package javafx;
```

```

public class CalculatorModel {
    private float calculationValue;

    public void calculate(float firstNumber, float secondNumber, char
operator) {
        switch (operator) {
            case '+':
                calculationValue = firstNumber + secondNumber;
                break;
            case '-':
                calculationValue = firstNumber - secondNumber;
                break;
            case '*':
                calculationValue = firstNumber * secondNumber;
                break;
            case '/':
                if (secondNumber != 0) {
                    calculationValue = firstNumber / secondNumber;
                } else {
                    throw new ArithmeticException("Cannot divide by
zero");
                }
                break;
            default:
                throw new IllegalArgumentException("Invalid operator");
        }
    }

    public float getCalculationValue() {
        return calculationValue;
    }
}

```

View:

```
package javafx;

import javafx.geometry.Insets;
import javafx.scene.control.Button;
import javafx.scene.control.Label;
import javafx.scene.control.TextField;
import javafx.scene.layout.BorderPane;
import javafx.scene.layout.GridPane;
import javafx.scene.layout.HBox;
import javafx.scene.layout.Priority;

public class CalculatorView {
    private TextField firstNumber, secondNumber, calcSolution;

    private Button addButton, subButton, mulButton, divButton;
    private BorderPane root;

    public CalculatorView() {
        firstNumber = new TextField();
        secondNumber = new TextField();
        calcSolution = new TextField();
        calcSolution.setEditable(false);

        addButton = new Button("+");
        subButton = new Button("-");
        mulButton = new Button("*");
        divButton = new Button("/");

        root = new BorderPane();
        root.setPadding(new Insets(12));

        GridPane infoPanel = new GridPane();
        infoPanel.setHgap(10);
        infoPanel.setVgap(10);

        infoPanel.add(new Label("First Number:"), 0, 0);
        infoPanel.add(firstNumber, 1, 0);
        infoPanel.add(new Label("Second Number:"), 0, 1);
        infoPanel.add(secondNumber, 1, 1);
        infoPanel.add(new Label("Result:"), 0, 2);
        infoPanel.add(calcSolution, 1, 2);

        HBox operandPanel = new HBox(10);
        operandPanel.getChildren().addAll(addButton, subButton, mulButton,
divButton);
        HBox.setHgrow(addButton, Priority.ALWAYS);
        HBox.setHgrow(subButton, Priority.ALWAYS);
        HBox.setHgrow(mulButton, Priority.ALWAYS);
        HBox.setHgrow(divButton, Priority.ALWAYS);
        addButton.setMaxWidth(Double.MAX_VALUE);
```

```

        subButton.setMaxWidth(Double.MAX_VALUE);
        mulButton.setMaxWidth(Double.MAX_VALUE);
        divButton.setMaxWidth(Double.MAX_VALUE);

        root.setCenter(infoPanel);
        root.setBottom(operandPanel);
    }

    public BorderPane getView() {
        return root;
    }

    public void addButtonListener(Runnable handlerAdd, Runnable handlerSub,
    Runnable handlerMul, Runnable handlerDiv) {
        addButton.setOnAction(e -> handlerAdd.run());
        subButton.setOnAction(e -> handlerSub.run());
        mulButton.setOnAction(e -> handlerMul.run());
        divButton.setOnAction(e -> handlerDiv.run());
    }

    public String getFirstNumber() {
        return firstNumber.getText();
    }

    public String getSecondNumber() {
        return secondNumber.getText();
    }

    public void setCalcSolution(String solution) {
        calcSolution.setText(solution);
    }
}
Controller:
package javafx;

public class CalculatorController {
    private CalculatorModel model;
    private CalculatorView view;

    public CalculatorController(CalculatorModel model, CalculatorView view) {
        this.model = model;
        this.view = view;

        view.addButtonListener(
            () -> calculate('+'),
            () -> calculate('-'),
            () -> calculate('*'),
            () -> calculate('/'));
    }

    private void calculate(char operator) {

```

```

        try {
            float firstNumber = Float.parseFloat(view.getFirstNumber());
            float secondNumber = Float.parseFloat(view.getSecondNumber());

            model.calculate(firstNumber, secondNumber, operator);

            view.setCalcSolution(String.valueOf(model.getCalculationValue()));
        } catch (NumberFormatException ex) {
            view.setCalcSolution("Error");
        } catch (ArithmeticException ex) {
            view.setCalcSolution("Error: Division by zero");
        } catch (IllegalArgumentException ex) {
            view.setCalcSolution("Error: Invalid operator");
        }
    }
}

Driver:
package javafx;

import javafx.application.Application;
import javafx.scene.Scene;
import javafx.stage.Stage;

public class MVCCalculator extends Application {

    @Override
    public void start(Stage stage) {
        CalculatorModel model = new CalculatorModel();
        CalculatorView view = new CalculatorView();
        new CalculatorController(model, view);

        Scene scene = new Scene(view.getView(), 420, 200);
        stage.setTitle("MVC Calculator (JavaFX)");
        stage.setScene(scene);
        stage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}

```

Kết quả chạy:

First Number: 1 Second Number: 2 Result: 3.0 + - * /	First Number: 6 Second Number: 0 Result: Error: Division by zero + - * /
---	---

3. Git

a. Add

```
endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[main]
% git add .
endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[main]
% git status
On branch main
Your branch is up to date with 'origin/main'.
```

Changes to be committed:

```
(use "git restore --staged <file>..." to unstage)
new file:   .gitignore
new file:   .vscode/launch.json
new file:   .vscode/settings.json
new file:   HelloWorld.java
new file:   MVC/javafx/CalculatorController.java
new file:   MVC/javafx/CalculatorModel.java
new file:   MVC/javafx/CalculatorView.java
new file:   MVC/javafx/MVCCalculator.java
new file:   MVC/swing/CalculatorController.java
new file:   MVC/swing/CalculatorModel.java
new file:   MVC/swing/CalculatorView.java
new file:   MVC/swing/MVCCalculator.java
```

b. Remove

```
endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[hw1-
hhoang] % git rm demo.txt
rm 'Homework01/20235729-PhanHuyHoang/demo.txt'
endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[hw1-
hhoang] % git commit -m "chore: removed demo.txt"
[hw1-hhoang 796bfaa] chore: removed demo.txt
1 file changed, 1 deletion(-)
delete mode 100644 Homework01/20235729-PhanHuyHoang/demo.txt
```

c. Modify

```
endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[hw1-
hhoang] % echo demo > demo.txt
endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[hw1-
hhoang] % git add demo.txt
```

```

endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[hw1-hhoang] % git commit -m "chore: added demo.txt for modification"
[hw1-hhoang 8359d3c] chore: added demo.txt for modification
1 file changed, 1 insertion(+)
create mode 100644 Homework01/20235729-PhanHuyHoang/demo.txt
endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[hw1-hhoang] % echo demo2 > demo.txt
endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[hw1-hhoang] % git add demo.txt
endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[hw1-hhoang] % git commit -m "chore: modified demo.txt"
[hw1-hhoang 3708a9f] chore: modified demo.txt
1 file changed, 1 insertion(+), 1 deletion(-)

```

d. Commit

```

endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[hw1-hhoang] % git commit -m "feat: Hoang added HW1"
[hw1-hhoang edccfa8] feat: Hoang added HW1
12 files changed, 377 insertions(+)
create mode 100644 Homework01/20235729-PhanHuyHoang/.gitignore
create mode 100644 Homework01/20235729-PhanHuyHoang/.vscode/launch.json
create mode 100644 Homework01/20235729-PhanHuyHoang/.vscode/settings.json
create mode 100644 Homework01/20235729-PhanHuyHoang/HelloWorld.java
create mode 100644 Homework01/20235729-PhanHuyHoang/MVC/javafx/CalculatorController.java
create mode 100644 Homework01/20235729-PhanHuyHoang/MVC/javafx/CalculatorModel.java
create mode 100644 Homework01/20235729-PhanHuyHoang/MVC/javafx/CalculatorView.java
create mode 100644 Homework01/20235729-PhanHuyHoang/MVC/javafx/MVCCalculator.java
create mode 100644 Homework01/20235729-PhanHuyHoang/MVC/swing/CalculatorController.java
create mode 100644 Homework01/20235729-PhanHuyHoang/MVC/swing/CalculatorModel.java
create mode 100644 Homework01/20235729-PhanHuyHoang/MVC/swing/CalculatorView.java
create mode 100644 Homework01/20235729-PhanHuyHoang/MVC/swing/MVCCalculator.java

```

e. Push

```

endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[hw1-hhoang] % git push origin hw1-hhoang
Enumerating objects: 33, done.
Counting objects: 100% (33/33), done.
Delta compression using up to 4 threads
Compressing objects: 100% (25/25), done.
Writing objects: 100% (31/31), 5.59 KiB | 2.79 MiB/s, done.

```

```
Total 31 (delta 8), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (8/8), completed with 1 local object.
remote:
remote: Create a pull request for 'hw1-hhoang' on GitHub by visiting:
remote:   https://github.com/ryujin05/2025.2-166155-06/pull/new/hw1-hhoang
remote:
To https://github.com/ryujin05/2025.2-166155-06.git
 * [new branch]      hw1-hhoang -> hw1-hhoang
```

f. Pull

```
endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[main]
% git pull
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 903 bytes | 903.00 KiB/s, done.
From https://github.com/ryujin05/2025.2-166155-06
   78c5fd6..2cdc085  main      -> origin/main
Updating 78c5fd6..2cdc085
Fast-forward
 Homework01/20235729-PhanHuyHoang/.gitignore      | 1 +
 Homework01/20235729-PhanHuyHoang/.vscode/launch.json | 31 +++
+++++++
 Homework01/20235729-PhanHuyHoang/.vscode/settings.json | 5 +++
++
 Homework01/20235729-PhanHuyHoang/HelloWorld.java | 15 +++
+++++++
 Homework01/20235729-PhanHuyHoang/MVC/javafx/CalculatorController.java | 34 +++
+++++++
 Homework01/20235729-PhanHuyHoang/MVC/javafx/CalculatorModel.java | 32 +++
+++++++
 Homework01/20235729-PhanHuyHoang/MVC/javafx/CalculatorView.java | 80 +++
+++++++
 Homework01/20235729-PhanHuyHoang/MVC/javafx/MVCCalculator.java | 25 +++
+++++++
 Homework01/20235729-PhanHuyHoang/MVC/swing/CalculatorController.java | 43 +++
+++++++
 Homework01/20235729-PhanHuyHoang/MVC/swing/CalculatorModel.java | 32 +++
+++++++
 Homework01/20235729-PhanHuyHoang/MVC/swing/CalculatorView.java | 68 +++
+++++++
 Homework01/20235729-PhanHuyHoang/MVC/swing/MVCCalculator.java | 11 +++
+++++++
12 files changed, 377 insertions(+)
create mode 100644 Homework01/20235729-PhanHuyHoang/.gitignore
create mode 100644 Homework01/20235729-PhanHuyHoang/.vscode/launch.json
create mode 100644 Homework01/20235729-PhanHuyHoang/.vscode/settings.json
create mode 100644 Homework01/20235729-PhanHuyHoang/HelloWorld.java
create mode 100644 Homework01/20235729-PhanHuyHoang/MVC/javafx/CalculatorController.java
```

```

create mode 100644
Homework01/20235729-PhanHuyHoang/MVC/javafx/CalculatorModel.java
create mode 100644
Homework01/20235729-PhanHuyHoang/MVC/javafx/CalculatorView.java
create mode 100644
Homework01/20235729-PhanHuyHoang/MVC/javafx/MVCCalculator.java
create mode 100644
Homework01/20235729-PhanHuyHoang/MVC/swing/CalculatorController.java
create mode 100644
Homework01/20235729-PhanHuyHoang/MVC/swing/CalculatorModel.java
create mode 100644
Homework01/20235729-PhanHuyHoang/MVC/swing/CalculatorView.java
create mode 100644
Homework01/20235729-PhanHuyHoang/MVC/swing/MVCCalculator.java

```

g. Merge

Hw1 hhoang #4

Merged phanhoang1366 merged 4 commits into `main` from `hw1-hhoang` now

Conversation 0 Commits 4 Checks 0 Files changed 12

phanhoang1366 commented now

quick merge

Mention @copilot in a comment to make changes to this pull request.

phanhoang1366 added 4 commits 6 minutes ago

- Feat: Hoang added Hw1 edccfa8
- chore: added demo.txt for modification 8359d3c
- chore: modified demo.txt 3708a9f
- chore: removed demo.txt 796bfaa

phanhoang1366 merged commit `2cdc085` into `main` now Revert

h. Branch

```

endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[main]
% git checkout -b hw1-hhoang
Switched to a new branch 'hw1-hhoang'
endeavour@inspiron143467 ..55-06/Homework01/20235729-PhanHuyHoang (git)-[hw1-
hhoang] % git branch

```

* hw1-hhoang
main